Cercis Algorithm (紫荆花算法)

Formal name: Cercis Self-Evolving Gatekeeper (CESG) **Location**: /g/Xiaogan_Supercomputing_data/SCX/theory/self_evolution/
Paper target: Nature Computational Science (Paper 2 of two-paper strategy)

Why "Cercis"

Cercis chinensis (紫荆) is a **cauliflorous** tree — its flowers bloom directly from old branches and trunks, not from new shoots.

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Cercis biological metaphor → SCX mathematical reality:

  Old branches & trunk     →  Memory bank M_t (accumulated structures, never delete
  Flowers                  →  Gatekeeper evaluations (new scores on old structures)
  Each spring bloom        →  Each iteration: S_t re-scores M_t
  Cauliflory               →  Resurrection: discarded structures "bloom" when S_t r
  New shoots               →  Novelty bonus  (t) • N_t(s) — exploration
  Deep roots               →  DFT/experimental ground truth (physical anchor)
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Tagline for paper: *"Knowledge does not grow from new data alone — it flowers from the old wood of accumulated experience."*

Paper 2: Cercis Flowers

Item	Detail
Title	*Cercis Flowers: A Self-Evolving Gatekeeper with Provable Convergence*
Core Theorems	Theorem SE-1 (Convergence), Theorem SE-2 (Completeness Bound)
Target	Nature Computational Science / Nature Machine Intelligence
Depends on	SCX framework (Paper 1, JMLR/TMLR)
Location	`theory/self_evolution/` (theorems), `paper/paper_gatekeeper/` (manuscript)